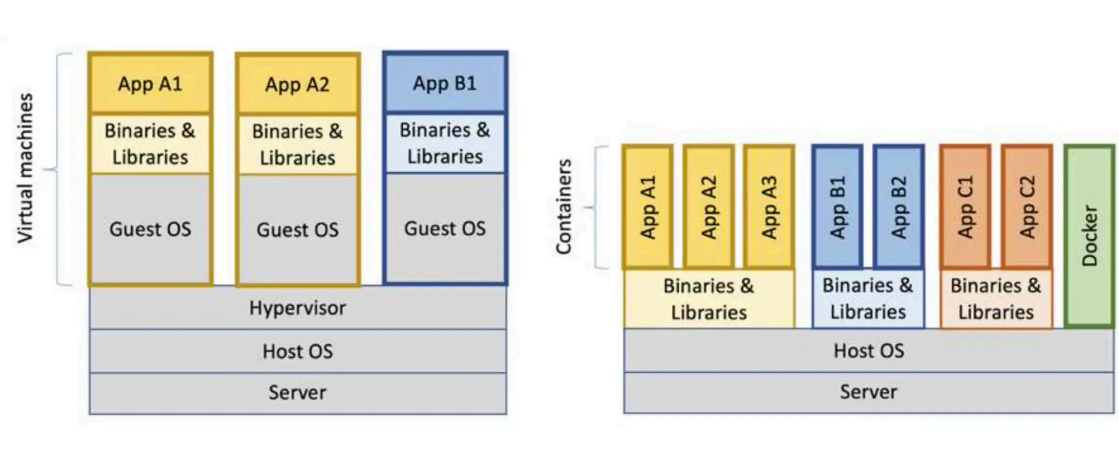
Introduction to Containerization

Deployment Models


Virtual machines
App A1
App A2
App B1
Binaries & Libraries
Binaries & Libraries
Binaries & Libraries
Guest OS
Guest OS
Guest OS
Hypervisor
Host OS
Server
Containers
App A1
App A2
App A3
App B1
App B2
App C1
App C2
Docker
Binaries & Libraries
Binaries & Libraries
Binaries & Libraries
Host OS
Server

- **Deployment density:** how efficiently workloads share the same infrastructure. Higher density leads to better resource utilization.
- **Isolation:** mechanisms that keep workloads separated (CPU, memory, filesystem, network), preventing interference and improving security and stability when many workloads run on the same infrastructure.

Bare Metal

- Apps run directly on physical servers
- Conflicts due to shared libraries & dependencies
- One-app-per-server → high cost, strong isolation
- Many-apps-per-server → low cost, low isolation

Virtual Machines

- Hypervisor (KVM, Xen, VMware ESXi) runs multiple VMs
- Each VM has its **own kernel and OS**
- OK density, strong isolation
- Drawback: memory & storage overhead, complex management

Containers

- Run on **host kernel** with **namespaces + cgroups**
- Package **only app + dependencies**
- Fast startup (<1s), minimal overhead
- High density, OK isolantion
- Perfect for **microservices and ephemeral workloads**

Chroot: Early Isolation

`chroot` changes a process root directory → “filesystem jail.”

Pros:

- Lightweight, easy to set up

Cons:

- Manual dependency management
- No resource control
- Breakable by root users

Example dependency inspection:

```
ldd /bin/ls
ldd /bin/bash
```

Example minimal root filesystem:

```
cage/
  bin/{bash, ls}
  usr/lib/{libc.so.6, libreadline.so.8, ...}
  lib64 -> usr/lib64
```

Run inside the jail:

```
sudo chroot cage /bin/ls
sudo chroot cage /bin/bash
```

Modern Isolation Primitives

Namespaces

- Kernel feature isolating system resources:
    - **Mount** → filesystem views
    - **PID** → isolated process tree
    - **Network** → private network stack
    - **IPC** → inter-process communication
    - **UTS** → hostname/domainname
    - **User** → UID/GID mapping

```
sudo unshare --mount --pid --fork --mount-proc bash
ps aux
```

Control Groups (cgroups)

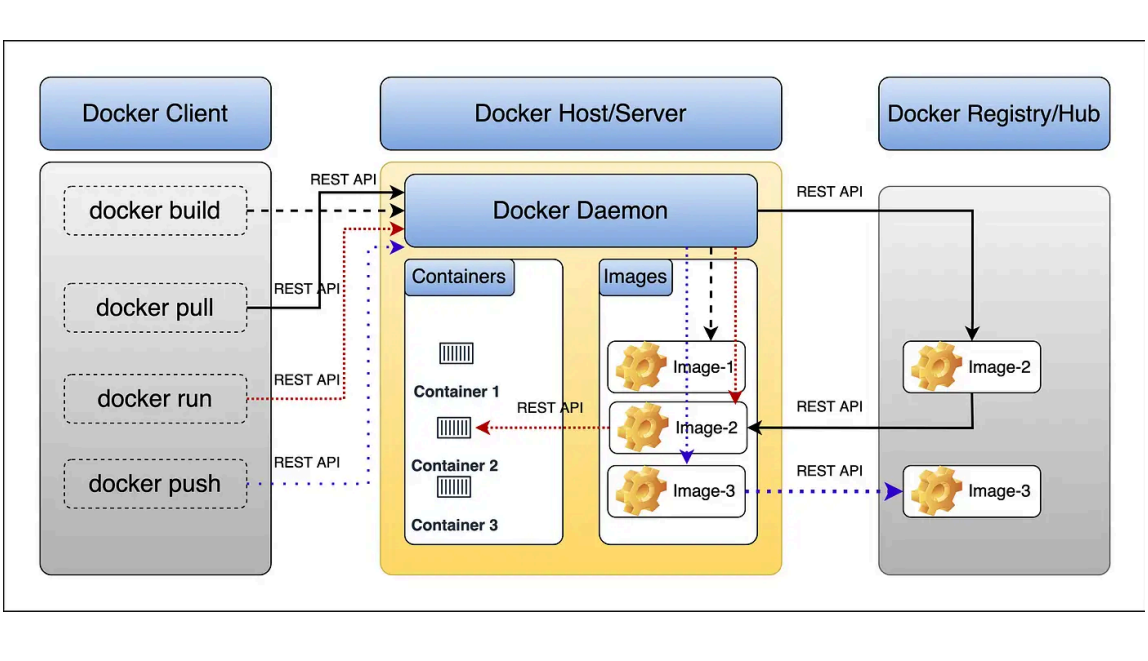
- Kernel mechanism to **limit & prioritize resources**:
    - CPU shares → fair CPU allocation
    - Memory limits → prevent OOMs
    - I/O throttling → control disk bandwidth
    - Process priority → scheduling weight

Container Technologies

Tool	Description	Use Case
systemd-nspawn	Lightweight OS-level containers, integrates with systemd	Testing, lightweight isolated environments
LXC / Incus	Full Linux OS containers, system-level isolation	Multi-service environments, lightweight VMs
Docker	Application containers, layered images, OCI-compliant	Running microservices, CI/CD pipelines
Podman	Docker-compatible, daemonless, supports rootless mode	Secure development and testing environments
Kubernetes	Container orchestrator: scheduling, scaling, networking	Multi-node deployments, production clusters

1. **LXC / LXD** → closest to a full OS, low-level containerization, more isolation, can run multiple services like a lightweight VM.
2. **systemd-nspawn** → lightweight system container, integrates with host systemd, good for sandboxed environments.
3. **Podman** → application-level containers, daemonless, rootless, very close to Docker but more secure.
4. **Docker** → application containers, layered images, widely used for microservices and CI/CD pipelines.
5. **Kubernetes** → orchestrates multiple containers across nodes; highest level of abstraction.

Docker Architecture


Docker Client
Docker Host/Server
Docker Registry/Hub
docker build
docker pull
docker run
docker push
REST API
REST API
REST API
REST API
Docker Daemon
REST API
Containers
Images
Container 1
Container 2
Container 3
REST API
Image-1
Image-2
Image-3
REST API
REST API
Image-2
Image-3

- Docker CLI → user interface
- Docker Engine/Daemon → build/run containers
- Images → immutable, versioned app bundles
- Containers → isolated runtime instances
- Registry → centralized storage

Essential Docker CLI

```
docker pull busybox
docker images
docker run -it busybox
docker ps
docker stop/start <id>
docker rm <id>
docker exec -it <id> bash
docker rmi busybox
```

Resources

- Docker Docs
- Top 8 Container Registries
- LXC/LXD Documentation
- Kernel Namespaces & cgroups